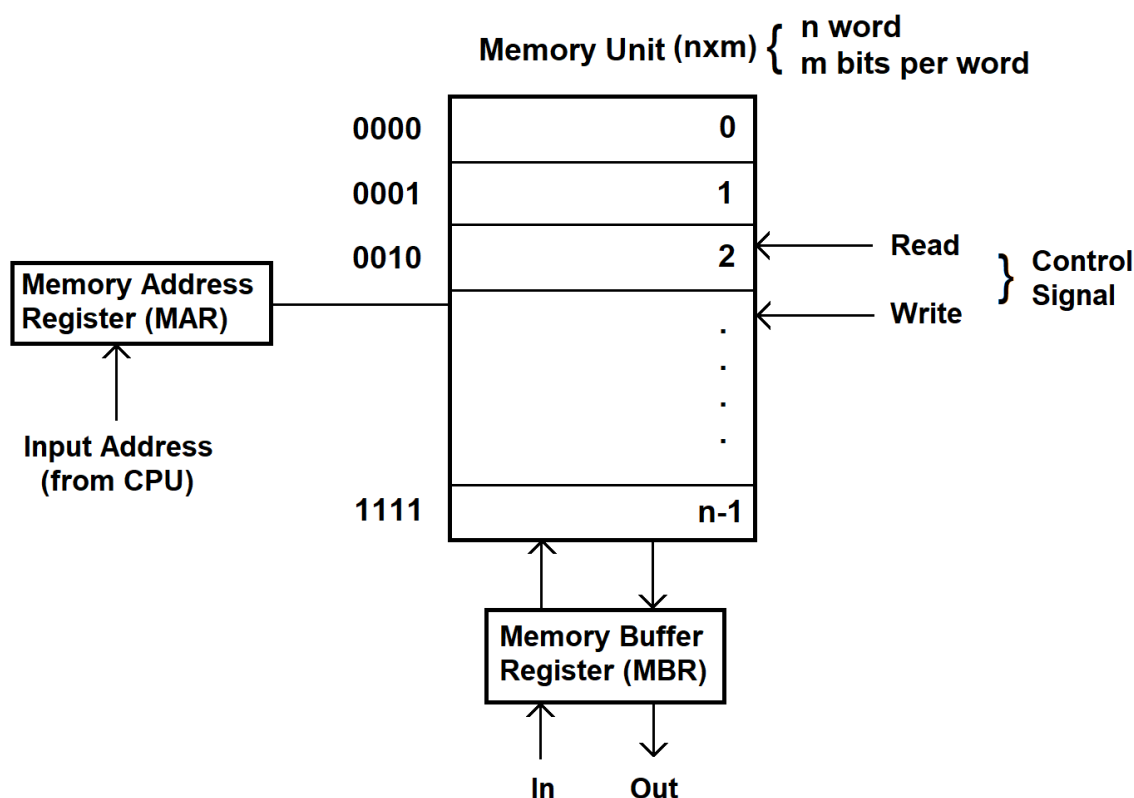


Lecture 11 - Memory



Memory Unit: A memory unit stores binary information in groups of bits called words.

Memory Address Register (MAR): To communicate with a memory word, its address is transferred to the memory address register. Internal circuit then opens the path to select that word. A n -bit address register can specify up to 2^n address.

Memory Buffer Register(MBR): Before writing and after reading information are kept in Memory Buffer registers. In each case, the address register specifies location of writing or reading.

Control Signal: The control lines Read and Write (usually combined into a single control line Read/Write) specifies the direction of transfer of the data.

Memory Enable	Read/Write	Memory Operation
0	X	None
1	0	Write to selected word
1	1	Read from selected word

Example: A memory unit of 1024 words with 8 bit/word needs, 10 address line (i.e. address register must contain 10 FF) and buffer register must have 8 FF to store words transferred in and out of it.

Memory Read Operation:

CPU issues a read request to the memory unit to fetch data from a specific memory location

Steps:

1. CPU sends the memory address from where data will be read into the MAR.
2. The memory unit activates the corresponding memory location.
3. The read control signal will be activated. (Read = 1 or RW = 0)
4. Data at the address is fetched and placed in the MBR.

Memory Write Operation:

CPU issues a write request to the memory to write data to a specific memory location

Steps:

1. CPU sends the memory address where data will be written into the MAR
2. Places the data to be written in the MBR
3. The write control signal will be activated. (Write = 1 or RW = 1)
4. Memory locates the given address and stores the data in it.

Formulas to remember:

What is the capacity of the memory?,

$$\text{Capacity} = \text{no. of word} \times (\text{bit/word})$$

How many cells are there?

$$\text{No. of word} = 2^{\text{(no. of address lines)}}$$

Sometimes, Number of Address lines is referred to only Address Lines

How many bits are required in MAR?

$$\text{Address lines} = \log_2(\text{No of word})$$

Unit Conversion:

8 bit = 1 byte

1024 byte = 1 KB

1024 KB = 1 MB

1024 MB = 1 GB

1024 GB = 1 TB

1024 TB = 1 PB

In short form, Bits are represented in small letter b. Example: 1 Kb = 1 Kilobits

And Bytes are represented in capital letter B. Example: 1 MB = 1 Megabyte

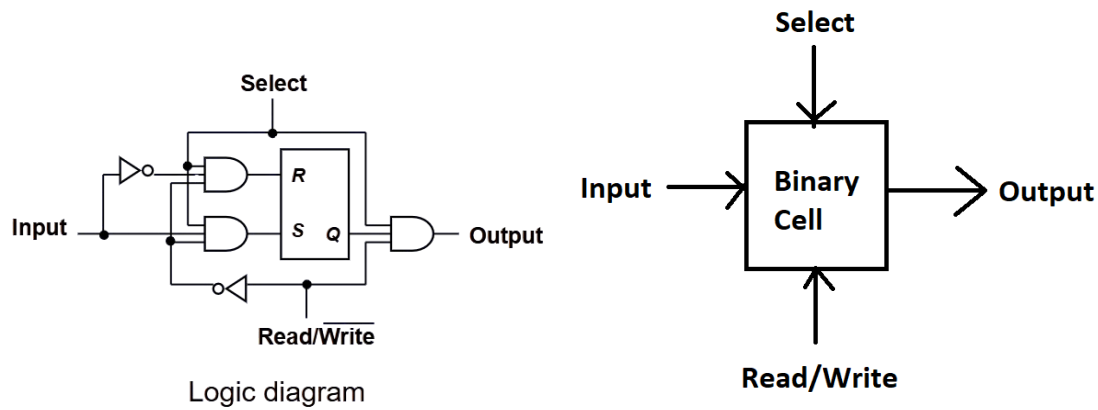
RAM: Random Access Memory

Two types of RAM:

Static RAMs use flip-flops as the memory cells. Complex circuit, No need to refresh.

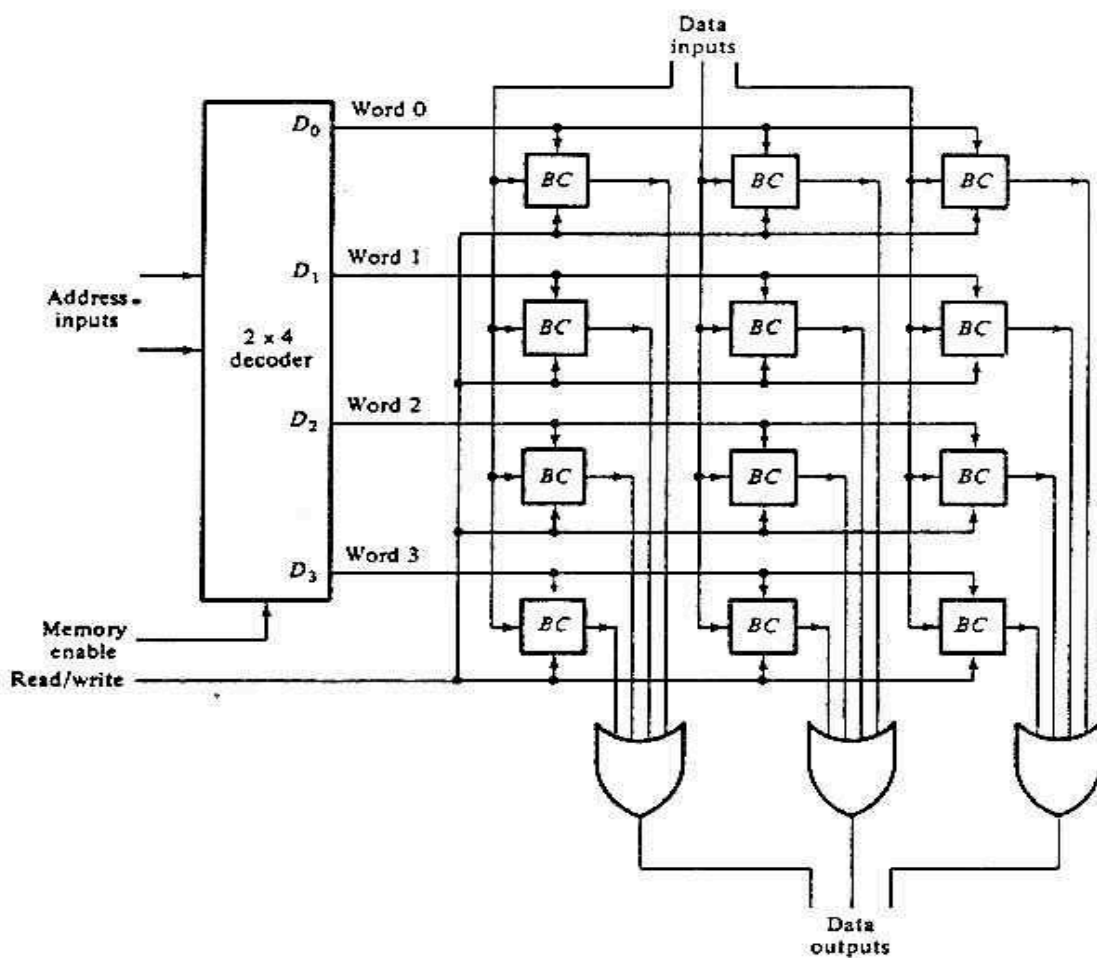
Dynamic RAMs use capacitor charges to represent data. Though simpler in circuitry, they have to be constantly refreshed.

Binary Cell: An unit of computer storage that can store one bit of information. A single memory cell of the static RAM has the following logic and block diagrams.



Logic construction of a 4 x 3 RAM (with decoder and OR gates):

Here 4 means 4 words. 3 means 3 bit per word



Practice Problems:

- How many address lines do we need for a 64 MB RAM with 16 bit/words?
- Design a 2048x32 RAM showing internal details.
- Design a 4096x16 RAM showing internal details.
- What is the capacity for a $2^{16} \times 16$ RAM in Gigabytes?
- Draw the block diagram of a binary cell.
- Draw and explain the functionalities of MAR and MBR.
- What type of information can you get from the “2048x32” part of a 2048x32 RAM?
- How many types of RAM are there?
- What is static RAM?
- What is dynamic RAM?
- Convert 8,796,093,022,208 Bits to Gigabytes.
- Convert 3 Terabytes to bits.